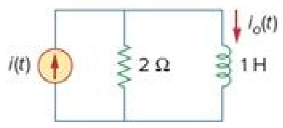
Problem

Obtain the current *io*(*t*) in the circuit of Fig. 18.44.

(a) Let *i*(*t*) = sgn(*t*) A.

(b) Let *i*(*t*) = 4[*u*(*t*) − *u*(*t* − 1)] A.

Figure 18.44



Step-by-step solution

Step 1 of 4

By current division, $$I_o = \frac{2}{2+j\omega}.I(\omega)$$

Step 2 of 4

(A) $for \quad i(t) = 5\,\text{sgn}(t)$

$$I(\omega) = \frac{10}{j\omega}$$

$$I_o = \frac{2}{2+j\omega}\bullet\frac{10}{j\omega} = \frac{20}{j\omega(2+j\omega)}$$

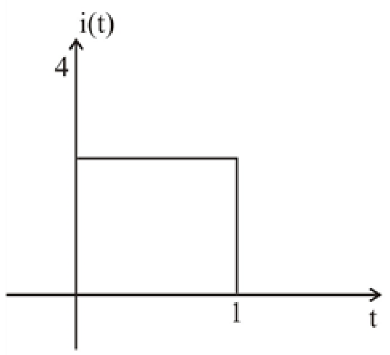
$$let \quad I_o = \frac{20}{s(s+2)} = \frac{A}{2s} + \frac{B}{s+2}, \quad A = 10, \qquad B = -10$$

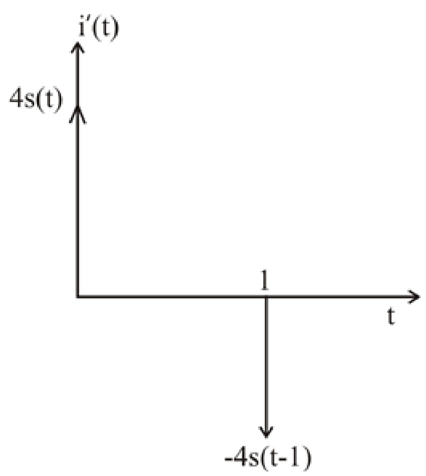
$$I_o(\omega) = \frac{10}{j\omega} - \frac{10}{2+j\omega}$$

$$i_o(t) = 5\,\text{sgn}(t) - 10e^{-2t}u(t)\,A$$

Step 3 of 4

(B)





$$i'(t) = 4s(t) - 4s(t-1)$$
$$j\omega I(\omega) = 4 - 4e^{-j\omega}$$
$$I(\omega) = \frac{4\left(1 - e^{-j\omega}\right)}{j\omega}$$
$$I_o \frac{8\left(1 - e^{-j\omega}\right)}{j\omega(2 + j\omega)} = 4\left(\frac{1}{j\omega} - \frac{1}{2 + j\omega}\right)\left(1 - e^{-j\omega}\right)$$
$$= \frac{4}{j\omega} - \frac{4}{2 + j\omega} - \frac{4e^{-j\omega}}{j\omega} + \frac{4e^{-j\omega}}{2 + j\omega}$$
$$i_o(t) = 2\,\mathrm{sgn}(t) - 2\,\mathrm{sgn}(t-1) - 4e^{-2t}u(t) + 4e^{-2(t-1)}u(t-1)\,A.$$